

Title: Plugging into the Future: An Exploration of Electricity Consumption Patterns

1. Problem Statement

India faces challenges in managing and distributing electricity efficiently due to varying consumption patterns across states. There is a lack of clear, visual insights into state-wise electricity usage, seasonal trends, and peak demand periods. This project aims to analyze electricity consumption data and present it through an interactive Tableau Story to help stakeholders make data-driven decisions.

2. Data Collection & Extraction

Data Source: Ministry of Power, Government of India, CEA reports

Time Period: 2019 - 2021

Parameters Collected: State name, Year, Month, Total electricity consumption in GWh, Per capita consumption

Tools Used: Excel for initial download, Web scraping for supplementary data

3. Data Preparation / Cleaning

Tool Used: Microsoft Excel & Tableau Prep

Cleaning Steps:

1. Removed duplicate entries and null values
2. Standardized state names and date formats
3. Handled missing data using average values for that state
4. Created calculated fields: `Year`, `Quarter`, `YoY Growth %`
5. Filtered out Union Territories with insufficient data

4. Data Visualization

Tool Used: Tableau Desktop

Chart Type | Purpose

Bar Chart | State-wise total consumption comparison

Line Chart | Year-on-year consumption trend

Map | Geographical distribution of electricity usage

5. Dashboard Design

Created 3 interactive dashboards:

- 1. Overview Dashboard:** Shows total consumption, top states, and national trend
- 2. State Analysis Dashboard:** Filter by state to see detailed monthly trends

Features Added: Filters for Year, State, Season. Tooltips for detailed info on hover.

6. Tableau Story

Combined all dashboards into a 6-point story:

- 1. Introduction** - Project objective
- 2. National Overview** - India's total consumption trend
- 3. Top Consumers** - States with highest usage
- 4. Growth Analysis**- YoY increase/decrease by state
- 5. Conclusion & Insights** - Key findings and recommendations

Published Link:

7. Results / Findings

1. Maharashtra, Gujarat, and Uttar Pradesh are the top 3 electricity consuming states
2. Peak consumption occurs during May-June due to summer demand
3. Southern states show steady growth of 4-6% YoY
4. Industrial states consume 3x more than agricultural states

8. Web Integration

The Tableau Story was embedded into a responsive website using Flask framework. The Arsha Bootstrap template was customized to display the project details, work process, and the interactive dashboard for public access.

9. Conclusion

This project successfully transforms raw electricity data into actionable insights. The interactive Tableau Story makes complex data accessible to policymakers and the general public. Future scope includes adding real-time data and forecasting models.

